

# Infinity Doc Creation 1— User Personas

## SafeNest: Building a Smart-Home Integration Hub

SafeNest: Your home—connected, protected, and perfectly in tune



SafeNest is an intelligent smart-home platform designed to simplify and secure daily life. It unifies smart devices under one seamless interface, offering homeowners personalized automation, proactive alerts, and effortless control, from climate to lighting to security. Built toward both convenience and peace of mind, SafeNest adapts to your routine, responds to real-world conditions, and helps you live better, simply.

## Doc 1—User Personas

### Background

SafeNest researchers and business analysts have been capturing market and customer data to understand the requirements for their new SafeNest home automation product.



# Questions

## Requirements identification—interview questions

Customers were asked 12 questions about their home automation needs:

- 01 Who lives in the home, and what are their key needs or constraints? (Ages, mobility, pets, visitors, tenants)
- 02 What outcomes from home automation matter most to you? (Convenience, safety, energy savings, caregiving, comfort)
- 03 What smart devices or platforms do you already use, and what frustrates you about them?
- 04 Walk me through a typical day at home. Where could automation remove steps or reduce worry?
- 05 Which rooms or systems are the highest priority to automate first, and why? (Lighting, climate, security, media, irrigation)
- 06 How do you prefer to control the home? (App, voice, physical switches, sensors, schedules) Who needs access and at what level?
- 07 What events should trigger alerts, and how do you want to be notified? (Phone push, SMS, email, siren) Do you need quiet hours?
- 08 What privacy and data handling expectations do you have? (Cameras, microphones, local versus cloud processing, logs, consent)
- 09 How important is reliability during internet or power outages? What offline behaviors or fallbacks do you expect?
- 10 Do you have any building or network constraints we should know about? (Rental rules, wiring, Wi-Fi coverage, panel space)
- 11 What budget and ongoing costs are acceptable? How will you judge return on investment (ROI) or value delivered?
- 12 How will you measure success after installation?

# Responses

A sample of user responses is listed below:

Tricia

01

Who lives in the home, and what are their key needs or constraints? (Ages, mobility, pets, visitors, tenants)

“I’m Tricia, 34, software engineer. I live with my partner and a very opinionated cat. We host friends often. Everyone needs the house to ‘just work’ without learning a new app every week. I’m happy to tinker; they want simple wall switches and reliable voice control.”

02

What outcomes from home automation matter most to you? (Convenience, safety, energy savings, caregiving, comfort)

“Customization and fine-grained control. I want scenes, automations, and routines that stack together and adapt to context. Convenience is big, but I also care about privacy and speed. Local first, cloud optional.”

03

What smart devices or platforms do you already use, and what frustrates you about them?

“I run Home Assistant on a mini-PC with a Zigbee coordinator. Devices include Hue lights, a few Shelly relays, Lutron Caséta dimmers, Aqara sensors, a Yale Assure lock, Ecobee thermostat, Sonos speakers, and a Rachio controller. Frustrations: vendor lock-in, cloud delays, and flaky integrations that drop after firmware updates.”

04

Walk me through a typical day at home. Where could automation remove steps or reduce worry?

“Morning: Bedroom lights fade up with a warm temperature, blinds open a bit, kettle outlet turns on. Workday: Presence detection switches my office scene, sets ‘do not disturb’ on speakers, and temp bands shift for one occupied room. Evening: ‘Movie’ scene dims lights, closes blinds, and pauses robot vacuum. Night: Lock doors, arm perimeter, and run a quiet HVAC curve. I want the hub to learn patterns and suggest tweaks, not just run static schedules.”

# Responses

Tricia, cont.

05

Which rooms or systems are the highest priority to automate first, and why? (Lighting, climate, security, media, irrigation)

“Lighting and climate in the main living spaces, then security and blinds. Lighting scenes change mood fast. Climate control is a big comfort win. After that, irrigation and garage, then media sync.”

06

How do you prefer to control the home? (App, voice, physical switches, sensors, schedules) Who needs access and at what level?

“I prefer automations and sensors as the default, with clean physical controls as the backup. Voice via Siri and Alexa is fine. I also want a single app with role-based access: full admin for me, simple scene controls for my partner, and guest mode via a QR code or NFC tag.”

07

What events should trigger alerts, and how do you want to be notified? (Phone push, SMS, email, siren) Do you need quiet hours?

“Critical only: water leak, door left unlocked, freezer temp rising, smoke/CO, and unusual motion when armed. Notifications via iOS push and Pushover; webhooks to a Slack channel for logs. Quiet hours 10 p.m.–7 a.m. with only safety alerts breaking through.”

08

What privacy and data handling expectations do you have? (Cameras, microphones, local versus cloud processing, logs, consent)

“Local processing preferred. Cameras should record to local storage with optional encrypted cloud backup. Logs stay on the LAN. I want granular consent for microphones and a transparent audit trail of automations and third-party access. No selling or sharing data.”

09

How important is reliability during internet or power outages? What offline behaviors or fallbacks do you expect?

“Very important. Automations must run locally without internet. Critical devices on a small UPS. If the hub goes down, wall switches and thermostats must still work. After outages, everything should restore state gracefully.”

# Responses

Tricia, cont.

10

Do you have any building or network constraints we should know about? (Rental rules, wiring, Wi-Fi coverage, panel space)

“Detached two-story house with a finished basement. I run UniFi networking with a separate IoT VLAN, strong Wi-Fi, and some Ethernet drops. No major wiring issues; many boxes have neutrals. Space in the rack for a hub and a small NAS.”

11

What budget and ongoing costs are acceptable? How will you judge return on investment (ROI) or value delivered?

“I’m fine with a one-time spend for solid hardware and a fair license if it adds real capability. Minimal monthly fees. ROI is fewer taps, faster response, happier housemates, and fewer ‘Is the door locked?’ moments.”

12

How will you measure success after installation?

“Subsecond light responses, stable routines with high uptime, reduced energy use, and zero ‘Why did the lights do that?’ surprises. I’d like dashboards that show time saved, comfort scores, and device health so I can tune without digging into logs.”

# Responses

Tricia, cont.

Specific compatibility and integration asks (Tricia volunteered):

“Matter over Thread preferred, with solid Zigbee 3.0 and Z-Wave Plus support.”

“Local LAN APIs, MQTT, webhooks, and clean Home Assistant integration. Node-RED or a rule engine with conditions, timers, and variables.”

“Works with Lutron Caséta/RA3, Philips Hue, Shelly, Aqara, Ecobee, Rachio, Sonos, Yale/Assure, and Nanoleaf. HomeKit, Alexa, and Google compatibility.”

“Scene layering, presence via Wi-Fi/Bluetooth and mmWave sensors, geofencing, NFC tags, and per-device OTA updates I can delay or roll back.”

# Responses

A sample of user responses is listed below:

Colin

01

Who lives in the home, and what are their key needs or constraints? (Ages, mobility, pets, visitors, tenants)

“I’m Colin, 42. My mom, Claire, is 78 and lives alone. She is steady most days but has arthritis and some short-term memory slips. She uses hearing aids. She wants to stay independent. She does not want to fuss with apps.”

02

What outcomes from home automation matter most to you? (Convenience, safety, energy savings, caregiving, comfort)

“Safety first. Early, accurate alerts without constant false alarms. Peace of mind for Mom and the family. It should support her dignity and routines, not take over her life.”

03

What smart devices or platforms do you already use, and what frustrates you about them?

“We have a Ring doorbell, two indoor cameras in common areas, and a basic medical alert pendant. Frustrations: too many apps, laggy notifications, and subscription creep. The pendant needs charging and she forgets to wear it. I want more passive safety that works in the background.”

04

Walk me through a typical day at home. Where could automation remove steps or reduce worry?

“She wakes at 7. Kitchen lights come on, kettle on, morning pills, light exercise. TV midmorning. Nap after lunch. Evenings are quiet. Risk points are stove being left on, bathroom slips, and a front door that sometimes stays unlocked. Night wandering is rare, but I worry. I want gentle prompts and automatic shutoffs before things become emergencies.”

# Responses

Colin, cont.

05

Which rooms or systems are the highest priority to automate first, and why? (Lighting, climate, security, media, irrigation)

“Bathroom safety and kitchen safety are top. Then, entry doors and bedroom. I want fall detection, bathroom occupancy checks, stove and oven safeguards, door-open alerts, and bed presence at night. After that, hallway lighting and hydration reminders.”

06

How do you prefer to control the home? (App, voice, physical switches, sensors, schedules) Who needs access and at what level?

“Mostly automatic. Mom should have simple wall switches and voice. No complex app for her. I want an app and a dashboard. My sister should get alerts but not admin rights. The visiting nurse needs a limited view and the ability to log wellness checks.”

07

What events should trigger alerts, and how do you want to be notified? (Phone push, SMS, email, siren) Do you need quiet hours?

“Critical alerts: detected fall, no movement for 2 hours during the day, bathroom occupancy over 30 minutes, stove on with no presence, water leak, exterior door opened at night, and low battery on safety devices. I want push notifications and SMS. If I do not respond in 2 minutes, escalate to my sister. For true emergencies, place an automated call. Quiet hours 10 p.m.–6 a.m., but safety alerts always break through. A daily morning digest would help.”

08

What privacy and data handling expectations do you have? (Cameras, microphones, local versus cloud processing, logs, consent)

“Cameras only in the entry and living room. No cameras in the bathroom or bedroom. Use motion, contact, and pressure sensors for private areas. Consent is key. Data shared only with family and the care team. Local storage for video and logs, with encrypted cloud backup for emergencies. Short retention and a clear audit trail.”



# Responses

Colin, cont.

09

How important is reliability during internet or power outages?  
What offline behaviors or fallbacks do you expect?

“Very important. Safety automations must run without internet. Battery backup for the hub, router, and key sensors. A cellular backup path for emergency calls and alerts. Physical panic buttons should still work. After an outage, the system should restore states and rearm by itself.”

10

Do you have any building or network constraints we should know about? (Rental rules, wiring, Wi-Fi coverage, panel space)

“Single-story bungalow from the 1960s. Some switch boxes do not have neutrals. Wi-Fi is fine in most rooms but weak near the back bathroom. We want minimal drilling and a clean look.”

11

What budget and ongoing costs are acceptable? How will you judge return on investment (ROI) or value delivered?

“I can fund a solid setup if it reduces risk. I am fine with a small monthly fee for cellular backup or monitored response if it adds real value. Value is fewer incidents, faster responses, and less stress for Mom and for us.”

12

How will you measure success after installation?

“Fewer false alarms. Fast, clear alerts when it matters. Mom reports feeling safe and not watched. We see simple trend views that show stable routines and early signs of issues.”

# Responses

Colin, cont.

Specific compatibility and integration asks (Colin volunteered):

“Passive fall detection using mmWave or camera-free sensing where possible. Option to integrate with Apple Watch fall detection or a pendant as backup.”

“Stove and oven safety: smart range shutoff or a sensor that cuts power after a no-presence timeout with a clear, audible prompt first.”

“Bathroom safety: humidity and occupancy sensors, floor leak sensor, and a soft night-light path to prevent trips.”

“Smart locks with caregiver codes, one-time EMS codes, and a way to unlock doors automatically if a verified emergency call is placed.”

“Medication reminders with a simple chime and a physical acknowledge button. Optional link to a dispenser.”

“Caregiver dashboard with activity trends, bathroom-visit patterns, sleep and bed-exit patterns, device health, and battery status. Exportable logs for the nurse.”

“Local processing, cellular failover, and role-based access. Minimal subscriptions. Clear printable instructions for Mom and a fridge card with a help button.”

# Responses

A sample of user responses is listed below:

Ellita

01

Who lives in the home, and what are their key needs or constraints? (Ages, mobility, pets, visitors, tenants)

“I’m Ellita, 38. I live with my partner and a very friendly dog. We garden, cook at home, and line-dry laundry in summer. I care about waste, noise, and light pollution. I am happy to learn a dashboard if it helps us save energy and water.”

02

What outcomes from home automation matter most to you? (Convenience, safety, energy savings, caregiving, comfort)

“Lower energy and water use without making life complicated. Maximize self-consumption of solar. Shift loads to cleaner grid times. Prevent leaks. I also care about products that are repairable and come in recyclable packaging from low-carbon manufacturers.”

03

What smart devices or platforms do you already use, and what frustrates you about them?

“We have a heat pump, a heat-pump water heater, smart thermostats, a drip irrigation controller, occupancy and leak sensors, and an EV charger. Frustrations: Each brand has its own app and the energy views do not line up. Some devices push cloud updates with no release notes. I want one hub that speaks energy.”

04

Walk me through a typical day at home. Where could automation remove steps or reduce worry?

“Morning is lights and coffee. Midday is a good time for laundry and dishwashing if the sun is strong. Evenings are cooking and showers. I want the hub to suggest the best time for high-draw tasks—precool in the afternoon if the grid will be dirty later, and irrigation paused if rain is due. At night, I want standbys trimmed.”

# Responses

Ellita, cont.

05

Which rooms or systems are the highest priority to automate first, and why? (Lighting, climate, security, media, irrigation)

“Water heating, space heating and cooling, EV charging, laundry, dishwasher, and irrigation. These have the biggest impact on use and bills. After that, window shades and ceiling fans to cut AC load and a hot water recirculation pump that only runs when someone is nearby.”

06

How do you prefer to control the home? (App, voice, physical switches, sensors, schedules) Who needs access and at what level?

“Mostly automatic with a clear ‘why’ shown on a card. I want a single app with three views: now, plan, and history. I get admin. My partner gets scene and schedule controls. Guests should only see lights and climate.”

07

What events should trigger alerts, and how do you want to be notified? (Phone push, SMS, email, siren) Do you need quiet hours?

“Water leak, abnormal continuous water flow, freezer temp rising, sump pump not cycling, inverter offline, and EV charging during a peak price window unless I approved it. Push notification first, then SMS if not acknowledged. Quiet hours 10 p.m.–6 a.m., safety alerts only.”

08

What privacy and data handling expectations do you have? (Cameras, microphones, local versus cloud processing, logs, consent)

“Local processing by default with optional encrypted cloud sync. No microphones always on. Keep detailed logs for 1 year, then summarize. Share nothing with third parties. I want a statement on embodied carbon and a take-back option for gear that reaches end of life.”

# Responses

Ellita, cont.

09

How important is reliability during internet or power outages?  
What offline behaviors or fallbacks do you expect?

“Important. Energy and water safety rules must work offline. If the grid is out, prioritize the fridge, sump, and network if we have a battery. After power returns, restore schedules and resync with the inverter.”

10

Do you have any building or network constraints we should know about? (Rental rules, wiring, Wi-Fi coverage, panel space)

“Small two-story home with good insulation and double glazing. Rooftop solar plus a movable ground array I reposition seasonally. Garden has two irrigation zones. Wi-Fi is strong, and I can add wired runs if needed. No gas service.”

11

What budget and ongoing costs are acceptable? How will you judge return on investment (ROI) or value delivered?

“I will pay for durable hardware and software that saves real kilowatt-hours and liters. Low monthly fees are fine if they provide useful forecasts or verified grid signals. Value is lower bills, lower CO<sub>2</sub>e, and less water use with the same comfort.”

12

How will you measure success after installation?

“Higher solar self-consumption, lower peak usage, lower total kWh per month, fewer irrigation hours with the same plant health, and a clear CO<sub>2</sub>e-avoided number. I want a simple scorecard and tips I can act on.”

# Responses

Ellita, cont.

Specific compatibility and integration asks (Ellita volunteered):

“Direct integration with my solar inverter and meter clamps for whole-home and circuit-level monitoring.”

“Forecast-driven scheduling that uses weather, solar prediction, and my utility’s time-of-use prices.”

“Grid carbon intensity and demand response support with automatic load shifting for water heating, EV charging, and laundry.”

“Smart irrigation that uses soil moisture, rain forecasts, and local bylaws. Skip cycles after rainfall and during freeze risk.”

“Window shades, fans, and thermostat coordination to precool or preheat when solar is plentiful.”

“Leak detection on mains and key fixtures, with a smart shutoff valve.”

“EV charger controls for off-peak and solar-only modes plus a manual ‘charge now’ override.”

“Right-to-repair details, spare parts availability, and a published support window for firmware. OTA updates I can review and delay.”

“Vendors that use recyclable packaging, publish environmental product declarations, and offer carbon-offset or low-carbon manufacturing with credible verification.”

“Local-first operation, open standards like Matter over Thread, and clean integrations with energy dashboards.”

# Tasks for Infinity

01

## Generate three user personas.

Take the customer interview data and generate user personas that characterize the customer types. We want a well-presented one-page persona for each user that features their image along with their name, a short bio, and demographics at the top of the page. Then three vertical sections listing: 1) Key Behaviors and Preferred Touchpoints; 2) Goals and Motivations featuring values, motivations, affinities, goals, and needs; and 3) Challenges and Frustrations, including demotivators. These sections are designed to better understand these users to help feature prioritization and release planning.

02

## Iterate on the personas to improve them.

Add privacy concerns and costs sensitivity.

Make sure the persona fits on one page.

Check that it is filling the page with useful detail (not too much to be cramped, not too little to have wasted space).

03

## Generate a new in-house stakeholder persona for internal audit and security requirements.